BANKING CLIENT ANALYTICS

Exploratory Data Analysis & Business Intelligence Report

Prepared by: **Data Analytics Team**
Dataset: **Banking Client Dataset | 3,000 Records | 25 Features**
Tools Used: **Python (Pandas, Seaborn, Matplotlib) | Power BI**
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PORTFOLIO PROJECT
This report is prepared as part of a data analytics portfolio showcasing EDA, statistical analysis, and business intelligence capabilities using a banking client dataset.

Executive Summary

This report presents a comprehensive Exploratory Data Analysis (EDA) of a banking client dataset comprising 3,000 client records and 25 distinct features. The analysis was conducted using Python (Pandas, NumPy, Seaborn, Matplotlib) and visualised through a Power BI interactive dashboard. The objective was to uncover behavioural patterns, financial segmentation insights, and risk profiles across the bank's diverse client base.

3,000 Total Clients *Unique records*	**25** Data Features *Across demographics & finance*	**5** Nationalities *European, Asian, American, Australian, African*

$171K Avg. Estimated Income *Range: $15K – $522K*	**$671K** Avg. Bank Deposits *Per client*	**$591K** Avg. Bank Loans *Per client*

Key Findings at a Glance

- Bank Deposits and Saving Accounts share the strongest positive correlation (0.75), indicating clients who save more also hold larger deposits.
- 49.2% of clients fall under the High fee structure, suggesting a premium-oriented client majority.
- 44.4% of clients are classified as Jade loyalty tier — the entry-level tier — indicating significant room for upselling.
- Risk Weighting 2 is the most common (40.7%), reflecting a predominantly low-to-moderate risk appetite across the client base.
- Estimated income moderately drives superannuation savings (r = 0.37), credit card balance (r = 0.30), and business lending (r = 0.33).
- European clients form the largest nationality segment (43.6%), followed by Asian (25.1%) and American (16.9%).

Business Problem Statement

Background & Context

Modern banks operate in an increasingly competitive environment where understanding the financial behaviour, loyalty patterns, and risk profiles of clients is critical for sustainable growth. Financial institutions collect vast amounts of client data — demographic information, account activity, loan history, credit usage, and savings — but this data often goes underutilised in strategic decision-making.

This project addresses a core challenge faced by retail and commercial banking units: how to transform raw client data into actionable intelligence that supports personalised services, risk management, and revenue optimisation.

Problem Statement

> ***"How can the bank leverage client demographic, financial, and behavioural data to segment its customer base, identify high-value client groups, assess financial risk exposure, and develop data-driven strategies that improve client retention, cross-sell financial products, and optimise fee and loyalty structures?"***

Business Objectives

The following objectives guided the exploratory data analysis:

- Understand the demographic composition of the client base across age, gender, nationality, and occupation segments.
- Analyse the distribution of income bands and financial product holdings (deposits, savings, loans, credit cards, business lending).
- Identify loyalty classification trends and their relationship with client financial behaviour and risk profiles.
- Examine fee structure allocation and assess whether premium segments are receiving adequate engagement.
- Quantify correlations between financial variables to reveal cross-sell and upsell opportunities.
- Profile clients by risk weighting to support the bank's risk management and credit policy teams.

Business Questions Addressed

#	Business Question	Analysis Type
1	What is the income distribution across the client base?	Univariate / Categorical
2	Which loyalty tier holds the most clients?	Univariate / Bar Analysis
3	How does nationality influence financial product usage?	Bivariate Analysis

#	Business Question	Analysis Type
4	Are high-income clients more likely to borrow?	Correlation Analysis
5	What is the risk profile of the client base?	Univariate / Risk Analysis
6	Do savings and deposits co-move significantly?	Correlation Heatmap
7	How are fee structures distributed?	Distribution Analysis
8	What is the age profile of the client base?	Numerical Analysis

Expected Business Outcomes

- Enable personalised marketing campaigns based on client income band and loyalty tier.
- Support risk teams with a clearer view of client risk weighting distribution.
- Identify under-served client segments (e.g., Jade-tier, low fee structure) that may benefit from targeted product offerings.
- Provide the foundation for predictive modelling (e.g., churn prediction, loan default risk) in subsequent project phases.

Dataset Overview

Dataset Summary

The dataset contains 3,000 client records collected from a retail and commercial banking environment. It encompasses 25 features spanning client demographics, financial product holdings, account activity, and segmentation identifiers.

Attribute	Details
Source	Banking Client Dataset (CSV Format)
Total Records	3,000 client entries
Total Features	25 columns
Missing Values	None detected — clean dataset
Data Types	Numerical (int, float) and Categorical (string)
Time Reference	Join date captured via 'Joined Bank' field

Feature Catalogue

Feature	Type	Description
Client ID	Categorical	Unique client identifier (e.g., IND81288)
Name	Categorical	Client full name
Age	Numerical	Client age (17–85 years)
Location ID	Numerical	Geographic location identifier
Joined Bank	Date	Date the client joined the bank
Banking Contact	Categorical	Mode of banking contact
Nationality	Categorical	Client nationality (5 categories)
Occupation	Categorical	Client occupation / job title
Fee Structure	Categorical	Fee tier: High / Mid / Low
Loyalty Classification	Categorical	Loyalty tier: Platinum / Gold / Silver / Jade
Estimated Income	Numerical	Annual estimated income ($)
Superannuation Savings	Numerical	Retirement savings balance ($)
Amount of Credit Cards	Numerical	Number of credit cards held
Credit Card Balance	Numerical	Total credit card balance ($)
Bank Loans	Numerical	Total bank loan amount ($)
Bank Deposits	Numerical	Total deposit balance ($)

Feature	Type	Description
Checking Accounts	Numerical	Checking account balance ($)
Saving Accounts	Numerical	Savings account balance ($)
Foreign Currency Account	Numerical	Foreign currency holdings ($)
Business Lending	Numerical	Business lending balance ($)
Properties Owned	Numerical	Number of properties owned
Risk Weighting	Ordinal	Risk score (1 = lowest, 5 = highest)
BRId	ID	Branch identifier
GenderId	Binary	Gender (1 = Male, 2 = Female)
IAId	ID	Internal account identifier

Methodology

Tools & Technologies

Tool / Library	Version	Purpose
Python	3.x	Core programming language
Pandas	Latest	Data loading, cleaning, manipulation
NumPy	Latest	Numerical computations
Seaborn	Latest	Statistical visualisations (histplots, countplots)
Matplotlib	Latest	Custom chart rendering and layout
Power BI	Desktop	Interactive dashboard and business reporting
Jupyter Notebook	Latest	Analytical workflow and code documentation

Analytical Workflow

Step 1 — Data Loading & Inspection

The dataset was loaded using pd.read_csv() and inspected using df.info(), df.describe(), df.columns, df.shape, and df.isnull().sum() to understand the structure, data types, and data quality.

Step 2 — Feature Engineering

A derived feature 'Income Band' was created using pd.cut() to segment continuous income values into three discrete groups:

- Low: Below $100,000
- Mid: $100,000 – $300,000
- High: Above $300,000

This binning enabled more meaningful categorical analysis of the income dimension.

Step 3 — Univariate Analysis

Value counts and frequency distributions were computed for all categorical and ordinal features including: BRId, GenderId, IAId, Amount of Credit Cards, Nationality, Occupation, Fee Structure, Loyalty Classification, Properties Owned, Risk Weighting, and Income Band. Count plots were rendered for visual comparison.

Step 4 — Bivariate Analysis

Cross-dimensional analysis was performed by plotting categorical features against Nationality as the hue variable, revealing how demographic segmentation maps onto product and behavioural features.

Step 5 — Numerical Distribution Analysis

Kernel Density Estimation (KDE) histograms were generated for eight numerical features: Age, Estimated Income, Superannuation Savings, Credit Card Balance, Bank Loans, Bank Deposits, Saving Accounts, and Business Lending. These revealed skewness, multimodality, and outlier patterns in the financial data.

Step 6 — Correlation Heatmap

A Pearson correlation matrix was computed for all numerical features and visualised as a heatmap using the 'crest' colour palette. This step was critical for identifying inter-feature relationships, informing product cross-sell opportunities and risk modelling considerations.

Step 7 — Power BI Dashboard

Key insights were translated into an interactive Power BI dashboard featuring client segmentation visuals, income band analysis, loyalty and fee tier distribution, and financial product overview panels.

Exploratory Data Analysis — Findings

5.1 Client Demographics

Age Distribution

The client age ranges from 17 to 85 years with a mean age of approximately 51 years. The distribution reflects a broad customer base, with a notable concentration in the 35–70 age bracket — typical for established retail banking clientele. Younger clients (17–30) represent a smaller but strategically important segment for long-term retention.

Gender Distribution

The gender split across the 3,000 clients is nearly equal: 50.4% female (GenderId = 2) and 49.6% male (GenderId = 1), suggesting no significant gender skew in the bank's client acquisition strategy.

Nationality Composition

Nationality	Count	Share (%)
European	1,309	43.6%
Asian	754	25.1%
American	507	16.9%
Australian	254	8.5%
African	176	5.9%

Europeans form the dominant client segment at 43.6%, followed by Asians at 25.1%. African clients represent the smallest segment (5.9%), presenting a potential opportunity for targeted outreach and growth.

5.2 Financial Segmentation

Income Band Distribution

The income band segmentation reveals that the majority of clients fall in the Mid income band ($100K–$300K), which aligns with the bank's predominantly high-fee-structure client base. A notable portion of clients are in the High income band (>$300K), indicating a significant wealth management opportunity.

Fee Structure Allocation

Fee Structure	Count	Share (%)
High	1,476	49.2%
Mid	962	32.1%
Low	562	18.7%

Nearly half of all clients (49.2%) are on the High fee structure. This concentration indicates that the bank serves a predominantly premium-tier clientele, though the 18.7% on Low fee structures may represent clients eligible for tier upgrades through targeted relationship management.

Loyalty Classification

Loyalty Tier	Count	Share (%)	Tier Level
Jade	1,331	44.4%	Entry
Silver	767	25.6%	Mid
Gold	585	19.5%	High
Platinum	317	10.6%	Premium

The loyalty pyramid shows a typical distribution with the largest population in the entry-level Jade tier (44.4%). Only 10.6% of clients have reached Platinum status — the bank's most valuable segment. This pyramid shape underlines opportunities for loyalty programme interventions to accelerate tier progression.

5.3 Financial Product Analysis

Key Financial Averages

$171K Avg. Estimated Income *Max: $522K*	**$672K** Avg. Bank Deposits *Strong savings behaviour*	**$591K** Avg. Bank Loans *High leverage ratio*

$3,176 Avg. Credit Card Balance *Relatively low balances*	**$25.5K** Avg. Superannuation *Retirement savings*	**17–85** Age Range *Mean age: 51 years*

Risk Weighting Profile

Risk Level	Count	Share (%)	Profile
1 — Very Low	836	27.9%	Conservative, low-risk clients
2 — Low	1,222	40.7%	Moderate savers, stable income
3 — Medium	460	15.3%	Moderate leverage, mixed portfolio
4 — High	322	10.7%	Higher borrowing, active credit use

Risk Level	Count	Share (%)	Profile
5 — Very High	160	5.3%	High exposure, elevated risk

Over 68% of clients are concentrated in risk levels 1 and 2, reflecting a predominantly conservative client base. This is broadly positive from a credit risk perspective but may suggest limited uptake of higher-yield financial products.

5.4 Correlation Analysis

Key Correlations from the Heatmap

Variable Pair	Correlation (r)	Interpretation
Bank Deposits ↔ Saving Accounts	0.75	Strong — high savers maintain high deposits
Business Lending ↔ Bank Loans	0.42	Moderate — businesses borrow actively
Business Lending ↔ Bank Deposits	0.44	Moderate — borrowers also maintain deposits
Estimated Income ↔ Superannuation	0.37	Moderate — higher earners save more for retirement
Estimated Income ↔ Business Lending	0.33	Moderate — higher income drives business borrowing
Estimated Income ↔ Credit Card Balance	0.30	Moderate — higher income correlates with card spend

The correlation analysis reveals that income is a key driver of both saving and borrowing behaviour. The strongest relationship — between deposits and savings (r = 0.75) — signals an integrated wealth management behaviour among the bank's clients and highlights the potential for bundled product offerings.

Business Insights & Recommendations

Insight 1 — Loyalty Tier Upsell Opportunity

44.4% of clients are in the Jade tier. The bank should implement a targeted loyalty progression programme with clear milestones, rewards, and personalised outreach to nudge Jade and Silver clients towards Gold and Platinum tiers.

Insight 2 — Income-Driven Cross-Sell Strategy

Since estimated income moderately correlates with superannuation savings, credit card balance, and business lending, the bank can use income band as a primary segmentation variable to tailor product recommendations. High-income clients should be prioritised for wealth management and superannuation products.

Insight 3 — Savings-Deposit Bundle Potential

The strongest correlation (r = 0.75) is between deposits and savings accounts. The bank should offer bundled savings-deposit products and promote automatic transfer mechanisms to deepen product penetration among existing savers.

Insight 4 — Fee Structure Upgrade Campaign

18.7% of clients on the Low fee structure likely qualify for Mid-tier products based on their income and financial activity. A proactive relationship management programme targeting these clients could increase revenue per client and improve retention.

Insight 5 — Risk-Based Product Tailoring

68.6% of clients have a risk weighting of 1 or 2 (conservative). These clients should be engaged with low-risk, stable-return investment products. The 5.3% with Risk Weighting 5 warrant close monitoring and may benefit from financial counselling or restructured credit facilities.

Insight 6 — Minority Nationality Outreach

African clients represent only 5.9% of the client base despite being a growing demographic. The bank should explore culturally relevant outreach programmes, multilingual support, and targeted products to expand this segment.

Conclusion

This project demonstrates the power of structured Exploratory Data Analysis in transforming raw banking data into strategic business intelligence. By applying Python-based data science techniques and Power BI visualisation, the analysis delivered a clear picture of client demographics, financial behaviour, product engagement, and risk profiles across 3,000 banking clients.

The findings reveal a bank with a largely conservative, European-dominant, and premium-fee client base that is heavily concentrated in entry-level loyalty tiers. The strong correlation between savings and deposits — alongside income-driven borrowing patterns — provides a clear quantitative foundation for personalised product strategies.

This project forms the foundation for future analytical work, including predictive modelling for client churn, loan default risk assessment, and customer lifetime value estimation. The combination of EDA and interactive dashboarding positions the bank to transition from reactive reporting to proactive, data-driven relationship management.

Project Deliverables

Banking.csv | BANKING_CASE_CODE.ipynb | Banking_Dashboard.pbix | This Report

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